A PROJECT REPORT ON

STATUS BOARD FOR PBL

SUBMITTED TO
MIT SCHOOL OF COMPUTING, LONI, PUNE IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE

BACHELOR OF TECHNOLOGY
(Computer Science & Engineering)

BY

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MIT School OF COMPUTING

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MIT SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

This is to certify that the Project report entitled

“STATUS BOARD FOR PBL”

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is a bonafide work carried out by him/her under the supervision of **Prof Sonali Deshpande**, and has been completed successfully.

External Guide

Seal/Stamp of the Company/College

Place :
Date :

DECLARATION

We, the team members

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Hereby declare that the project work incorporated in the present project entitled **“Status Board for PBL”** is original work. This work (in part or in full) has not been submitted to any University for the award or a Degree or a Diploma. We have properly acknowledged the material collected from secondary sources wherever required. We solely own the responsibility for the originality of the entire content.

Date: 19/07/2021

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EXAMINER’S APPROVAL CERTIFICATE

The project report entitled “**STATUS BOARD FOR PBL**” submitted by Nirmiti Kulkarni (ADT24SOCB0691), Shreya Tamore (ADT24SOCB1122), Soham Hudge (ADT24SOCB1182), Soumyaranjan Parida (ADT24SOCB1207)

in partial fulfillment for the award of the degree of Bachelor of Technology (Computer Science & Engineering) during the academic year 2025-26, of MIT-ADT University, MIT School OF COMPUTING, Pune, is hereby approved.

Examiners:

1.

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ABSTRACT

In academic institutions, tracking project progress across multiple student groups and supervising faculty often becomes challenging due to unorganized communication and manual record-keeping. The **Status Board** project aims to address this issue by developing a digital project-tracking system that introduces a clear hierarchical workflow connecting **Heads → Faculty → Student Groups**.

The system allows faculty to create and manage project groups, set deadlines, and monitor progress, while Heads can oversee faculty performance and overall institutional progress in real time. Student groups can update their status, upload reports, and view assigned tasks or deadlines on a unified dashboard.

This structured platform enhances transparency, accountability, and efficiency in project supervision. It minimizes administrative delays, ensures timely submissions, and provides insightful progress analytics to institutional authorities. The Status Board thus acts as a comprehensive tool to streamline academic project management through an accessible and user-friendly interface.

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Chapter 1 INTRODUCTION

Project-based learning is an essential component of academic curriculums in many educational institutions, requiring students to work in groups, coordinate with faculty guides, and complete projects within defined timelines. However, as the number of student groups and supervising faculty increases, managing and tracking project progress becomes challenging. Traditional methods such as manual registers, spreadsheets, or basic digital tools often result in communication gaps, delayed submissions, and lack of real-time visibility.

To address these challenges, the **Status Board** project proposes a centralized, digital project-tracking system designed specifically for institutional workflows. The system introduces a structured hierarchy connecting **Heads**, **Faculty**, and **Student Groups**, enabling smooth coordination and efficient supervision. Faculty members can set deadlines, assign tasks, review submissions, and monitor progress, while Heads can oversee faculty performance and obtain consolidated insights into institutional project activities.

By providing a user-friendly dashboard, automated status updates, and role-based access control, the Status Board aims to enhance transparency, accountability, and productivity in academic project management. The system not only streamlines the tracking process but also reduces administrative overhead, making it easier for institutions to evaluate student performance and maintain project-related records effectively.

Existing Work

In many educational institutions, project tracking is carried out using traditional or partially digital methods, each with its own challenges and limitations. The most common existing systems are manual registers, spreadsheets, Learning Management Systems (LMS), and generic project management tools.

The **manual project tracking method** involves maintaining physical registers or paper-based records of project progress. Although this method is simple and requires no technological setup, it is highly time-consuming, prone to misplacement or damage, and does not offer real-time visibility. Managing multiple student groups becomes difficult, and faculty often struggle to maintain updated records.

Some institutions rely on **spreadsheets and shared drives**, such as Google Sheets or Excel files. While these tools allow basic collaboration and are easy to update, they lack role-based access control, require manual entry, and do not scale effectively when dealing with a large number of student groups. They also lack automation features like notifications or analytics.

Many colleges use **Learning Management Systems (LMS)** such as Google Classroom, Moodle, or Canvas for assignment submissions and communication. These platforms are useful for uploading files and sharing instructions; however, they are not designed specifically for project tracking. They do not support hierarchical supervision between Heads, Faculty, and Students, and offer limited customization for project workflows.

In some cases, faculty and students adopt **generic project management tools** such as Trello, Asana, or Microsoft Teams. These applications provide task boards, deadline reminders, and collaboration features, but they are not tailored for academic needs. They do not include institutional roles, data privacy controls, or academic evaluation features. Many advanced features also require paid subscriptions.

A few institutions have developed **internal project portals**, but these systems tend to be department-specific, limited in functionality, and lack scalability. They may also offer minimal analytics and outdated interfaces.

Overall, the existing work in project tracking systems shows significant gaps in centralization, communication, hierarchical monitoring, and progress visualization. These limitations highlight the need for a dedicated, institution-specific system like the **Status Board**, which provides real-time tracking, structured supervision, and better control over project workflows.

Motivation

In the modern digital era, technology has become an inseparable part of every sector, transforming how people work, learn, and communicate. Despite this digital revolution, many educational institutions and organizations still rely on **manual systems** to manage projects, assign tasks, and monitor progress. These traditional methods—such as using spreadsheets, physical records, or verbal communication—often lead to **data duplication, delays, miscommunication, and poor coordination** among team members. As the number of projects and participants increases, managing everything manually becomes time-consuming and inefficient.

The **motivation** behind developing a digitalized system stems from the growing need to overcome these challenges through automation and real-time collaboration. Digitalization in the labour market has already demonstrated how technology can streamline workflows, improve productivity, and ensure transparency. Applying the same approach to academic environments can significantly enhance the way institutions handle project-based learning. A centralized digital system allows administrators, faculty, and students to interact efficiently, reducing confusion and ensuring that progress is tracked continuously.

For faculty members and project heads, digitalization provides a **smart and centralized platform** to assign projects, monitor deadlines, and assess student performance from one interface. It eliminates repetitive manual entries and offers instant updates on group and individual progress. For students, it provides better clarity on assigned tasks, encourages accountability, and enables seamless communication with mentors. Such a system not only improves task management but also develops digital discipline and responsibility among students—skills essential in today's evolving job market.

Furthermore, digital transformation supports **eco-friendly and sustainable practices** by reducing the dependence on paper documentation and physical storage. It promotes the concept of **smart campuses and digital workplaces**, aligning with the government's vision of a digitally empowered society. By introducing automation and structured workflows, institutions can save time, reduce human error, and create a transparent environment for learning and evaluation.

Ultimately, the motivation behind this project lies in creating an **efficient, reliable, and collaborative digital ecosystem** that replaces traditional manual systems. It aims to bridge the gap between technology and education, ensuring that project tracking, communication, and evaluation become smoother, faster, and more accurate. Through this digital initiative, institutions can build a foundation for innovation, efficiency, and excellence in the way academic projects are managed and executed.

Objective and Scope

Objectives

The primary objective of the **Smart Task Management System** is to develop a centralized, role-based platform that simplifies and automates the management of academic projects within educational institutions. The system aims to provide an efficient and transparent way for **Heads, Faculty, and Students** to manage, track, and monitor project-related tasks in real time. Key objectives include:

1. **Automation of Project Tracking:** To eliminate manual record-keeping and reduce duplication of data through digital dashboards and automated updates.

2. **Efficient Communication:** To establish seamless communication channels between students, faculty, and department heads through built-in chat and feedback modules.

3. **Transparency and Accountability:** To ensure that every task, submission, and evaluation is recorded systematically for easy monitoring and review.

4. **Time Management:** To enhance productivity by enabling task scheduling, setting deadlines, and generating automated progress notifications.

5. **Centralized Data Storage:** To maintain secure and easily accessible project data for all departments.

By achieving these objectives, the system will enhance coordination, minimize delays, and ensure that academic projects progress smoothly toward completion.

Scope

The scope of this system extends across multiple departments within an educational institution. It is designed to handle various projects, regardless of team size or domain, providing **scalable and flexible functionality** for both undergraduate and postgraduate programs. The system can be implemented institution-wide to provide:

- **Multi-Department Integration:** Allowing simultaneous management of multiple projects under different faculties.

- **Real-Time Monitoring:** Heads and faculty can track project progress, generate reports, and make data-driven decisions.

- **User Role Customization:** Different user roles (Head, Faculty, Student) ensure personalized access and permissions for efficient workflow.

- **Data Security & Reliability:** Ensuring that stored data remains secure, backed up, and accessible only to authorized users.

- **Scalable Architecture:** Designed to accommodate future expansion, including AI-based analytics or integration with Learning Management Systems (LMS).

Overall, the Smart Task Management System provides a **feasible, cost-effective, and innovative solution** to digitalize project tracking and communication in academic settings, promoting transparency, accountability, and efficiency.

Chapter 2 CONCEPTS AND METHODS

The **Status Board** project is based on a combination of **software engineering principles**, **project management techniques**, and **database-driven web application design**. The following key concepts and methods were applied during its development:

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1. Software Engineering Concepts

- **Modular Design:**
  The system was divided into independent modules — Head, Faculty, and Student — to simplify development and maintenance.

- **Structured Development Lifecycle:**
  Followed the **Waterfall Model** of SDLC, ensuring systematic progress through requirement analysis, design, implementation, testing, and deployment.

- **Requirement Specification (SRS):**
  Prepared detailed documentation defining functional and non-functional requirements before coding.

- **Testing and Validation:**
  Conducted unit, integration, and user acceptance testing to ensure quality and reliability.

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2. Database and Backend Concepts

- **Relational Database Management System (RDBMS):**
  Used **MySQL** to store user details, project information, deadlines, and status updates.

- **Entity Relationship (ER) Modeling:**
  Designed ER diagrams to define relationships between different entities (Head, Faculty, Student, Projects).

- **Data Normalization:**
  Implemented normalization to eliminate redundancy and improve data consistency.

- **CRUD Operations:**
  Created, Read, Update, and Delete operations were used for managing project data efficiently.

3. Web Development Methods

- **Frontend Design:**
  Used **HTML**, **CSS**, and **JavaScript** to create a responsive and user-friendly interface.

- **Backend Development:**
  Implemented server-side logic using **PHP** (or Node.js/Python, depending on chosen stack).

- **Server Communication:**
  Employed HTTP methods (GET, POST) to handle data exchange between client and server.

- **Authentication and Access Control:**
  Incorporated login validation and role-based access to ensure secure data handling.

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4. Project Management Concepts

- **Hierarchical Workflow:**
  Designed a structured hierarchy of users (Heads → Faculty → Student Groups) for clear accountability.

- **Task Scheduling and Deadlines:**
  Faculty can assign project milestones with deadlines to student groups.

- **Progress Tracking:**
  Real-time dashboards display the current status of each project for both faculty and heads.

- **Version Control:**
  Used **GitHub** for source code management and collaboration among developers.

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5. User Experience and Interface Design

- **Dashboard Visualization:**
  Designed intuitive dashboards to display tasks, deadlines, and progress in an easy-to-read format.

- **Responsive Layout:**
  Ensured compatibility across devices using modern CSS frameworks.

- **Simplified Navigation:**
  Clear menus and forms for each user type minimize confusion and enhance usability.

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6. Security and Data Integrity

- **User Authentication:**
  Implemented secure login system to protect user data.
- **Data Encryption:**
  Passwords and sensitive data stored securely in the database.
- **Access Restriction:**
  Each role has limited access rights based on privileges (Head, Faculty, Student).

Literature Survey

In recent years, digital transformation in educational institutions has significantly influenced how academic and project-based activities are managed. Traditionally, project tracking and task management within colleges were handled using manual methods such as registers, spreadsheets, or Google Sheets. While these tools provided a basic structure, they lacked automation, real-time updates, and centralized data control. As a result, managing multiple student groups, monitoring project timelines, and ensuring consistent communication among faculty and students became increasingly difficult.

According to various studies and institutional reports, manual tracking often results in challenges such as **data duplication, delayed task updates, loss of accountability, and communication gaps** between project stakeholders. Research on academic management systems (Sharma & Gupta, 2021; Kumar et al., 2022) emphasizes the need for digital platforms that streamline project tracking, automate updates, and provide role-based access for different users such as heads, faculty, and students. Existing systems like Trello, Asana, and Monday.com have proven effective in corporate environments, yet they are not fully adapted to the academic ecosystem, where multiple teams, departments, and evaluation criteria must be managed simultaneously.

Several universities have experimented with customized task management portals or Learning Management Systems (LMS), but these often focus on course delivery rather than detailed project tracking. In a study by Singh and Mehta (2023), it was noted that while LMS platforms enhance communication, they lack fine-grained control over progress tracking, task assignment, and performance analytics in academic projects. Therefore, there is a growing demand for a **specialized smart task management system** tailored to academic institutions, integrating both task automation and collaborative features.

The proposed “Smart Task Management System” bridges this gap by combining automation, real-time monitoring, and role-based dashboards. It enables **faculty members to assign, track, and evaluate tasks**, while allowing **students to update progress and communicate effectively**. Heads or administrators can access centralized dashboards for overall project visibility, ensuring transparency and timely decision-making.

Chapter 3 PROJECT PLAN

1. Requirement Analysis

In this initial phase, detailed discussions were conducted with institutional staff, faculty members, and students to understand the existing challenges in project tracking.
Key requirements identified included:

- Role-based access for Heads, Faculty, and Students.
- Ability for Faculty to set deadlines and monitor submissions.
- Dashboard visualization for progress tracking.
- Secure authentication and data management.

A **Software Requirement Specification (SRS)** document was prepared to define all functional and non-functional requirements precisely.

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2. System Design

Based on the gathered requirements, the overall architecture of the system was designed.

- The **Head Module** was designed to manage faculty and view project statistics.
- The **Faculty Module** allows deadline setting, progress checking, and student feedback.
- The **Student Module** provides submission upload, task updates, and progress views.
  System design diagrams such as **DFD (Data Flow Diagram)** and **ER (Entity Relationship)** diagrams were created to visualize the data flow and database schema.

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3. Implementation

In this phase, the coding and module development were carried out.

- The front end was developed using **HTML, CSS, and JavaScript** for user interface design.
- The back end was developed using **PHP/Python/Node.js** (depending on the chosen stack) with **MySQL** or **Firebase** for database management.
- Each module (Head, Faculty, Student) was implemented separately and integrated after individual testing.

Version control tools such as **GitHub** were used for code management and collaboration.

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4. Testing

Once implementation was complete, the system underwent thorough testing to ensure accuracy, stability, and usability.

- **Unit Testing** was performed on each module to verify individual functionality.
- **Integration Testing** ensured smooth communication between modules.
- **User Acceptance Testing (UAT)** was conducted with faculty and student groups to gather feedback.

All identified bugs were fixed, and the final version of the Status Board was validated for deployment.

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5. Deployment and Maintenance

After successful testing, the system was deployed on a local server for institutional use.

User credentials were assigned, and a demonstration session was conducted for Heads, Faculty, and Students.

Post-deployment, minor feedback-based updates were carried out to improve usability. Future maintenance will focus on adding new features such as graphical analytics and mobile accessibility.

Chapter 4 SOFTWARE REQUIREMENT SPECIFICATION

1.1 Purpose of This Document

This SRS describes the functional and non-functional requirements for the **Status Board** system. It provides a detailed specification of features, constraints, interfaces, and acceptance criteria to guide design, development, testing, and deployment.

1.2 Scope of the System

Status Board is a web-based project-tracking system for educational institutions. It implements a role-based hierarchical workflow (Heads → Faculty → Student Groups) enabling: group creation and management, deadline assignment, progress updates, file submissions, notifications, and reporting. The system targets internal institutional usage to improve visibility and accountability in academic project management.

1.3 Definitions, Acronyms and Abbreviations

HOD / Head — Head of Department or equivalent authority.

Faculty — Project supervisor/teacher.

Student Group — One or more students working on a project.

Admin — System administrator (account and system settings).

Chapter 5 RESULTS

After implementing and testing the Status Board: A Project Tracking System for an Institution, the following results and observations were recorded based on system performance, user feedback, and practical usability within a small institutional setup:

1. Successful Hierarchical Workflow:
   The system effectively established a clear chain of access — from Heads to Faculty to Student Groups. Each level could view and manage relevant data without overlap or confusion.

2. Smooth Functionality of Dashboards:
   All user dashboards (Head, Faculty, and Student) performed as intended, displaying live status updates, assigned tasks, and project deadlines accurately.

3. Improved Monitoring and Coordination:
   Faculty members reported improved visibility into student progress and deadlines, which helped in timely feedback and evaluation. Heads were able to oversee faculty activities efficiently.

4. Enhanced Communication and Transparency:
   Communication gaps between faculty and students were minimized. The progress tracking system ensured that all updates and changes were visible in real time.

5. Timely Project Submissions:
   Automated reminders and structured deadlines led to more punctual submissions from student groups during testing phases.

6. User Feedback:
   Users appreciated the simplicity and intuitiveness of the interface. Minor suggestions included adding a mobile-friendly version and graphical performance summaries.

Chapter 6 CONCLUSION AND FUTURE WORK

The project **“Status Board: A Project Tracking System for an Institution”** successfully achieved its objective of creating an efficient, transparent, and user-friendly platform for managing academic projects within an institutional hierarchy.

Through the implementation of structured modules for Heads, Faculty, and Student Groups, the system streamlined communication, simplified supervision, and improved project visibility. Faculty members could easily monitor group progress and set deadlines, while Heads gained a centralized view of institutional performance.

The project demonstrated how digital project tracking can reduce manual errors, enhance accountability, and save time for both students and faculty. Overall, the **Status Board** proved to be a reliable and scalable system that contributes to improved institutional management and workflow efficiency.

Future Work

While the current version of the **Status Board** fulfills its core objectives, there are several areas for enhancement and expansion in future iterations:

1. **Mobile Application Integration:**
   Develop an Android/iOS version for improved accessibility and real-time updates.

2. **Automated Notifications:**
   Implement email or SMS alerts for task reminders, deadline updates, and feedback.

3. **Analytics Dashboard:**
   Add graphical insights and performance analytics to help Heads and Faculty evaluate progress trends.

4. **Role Expansion:**
   Introduce additional roles such as external evaluators or coordinators for broader project review and assessment.

5. **Cloud Deployment:**
   Shift the system to a cloud-based environment for improved scalability, data backup, and multi-campus support.

Chapter 7 BIBLIOGRAPHY

[1] Pressman, R. S., & Maxim, B. R. (2019). *Software Engineering: A Practitioner's Approach*. McGraw-Hill Education.

[2] Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson Education.

[3] Ian Sommerville, "Requirements Engineering," *Software Engineering*, Pearson, 2016.

[4] M. Fowler, "UML Distilled: A Brief Guide to the Standard Object Modeling Language," Addison-Wesley, 2004.

[5] W3Schools. *HTML, CSS, JavaScript Tutorials.* Retrieved from https://www.w3schools.com

[6] GeeksforGeeks. *Database Management and Web Development Resources.* Retrieved from https://www.geeksforgeeks.org

[7] Research Paper: Elbasha, A. M., & Abdellatif, M. M. (2025). *AIoT-based Smart Traffic Management System.* arXiv preprint.

[8] Documentation: *MySQL Developer Guide.* Oracle, 2024.

[9] Tutorialspoint. *Software Development Life Cycle (SDLC) – Models and Phases.* Retrieved from https://www.tutorialspoint.com

[10] GitHub Docs. *Version Control and Project Collaboration Guide.* Retrieved from https://docs.github.com